

Computational Microelectronics

Lecture 4 (Pre-recorded)

Sung-Min Hong (smhong@gist.ac.kr)

Semiconductor Device Simulation Laboratory

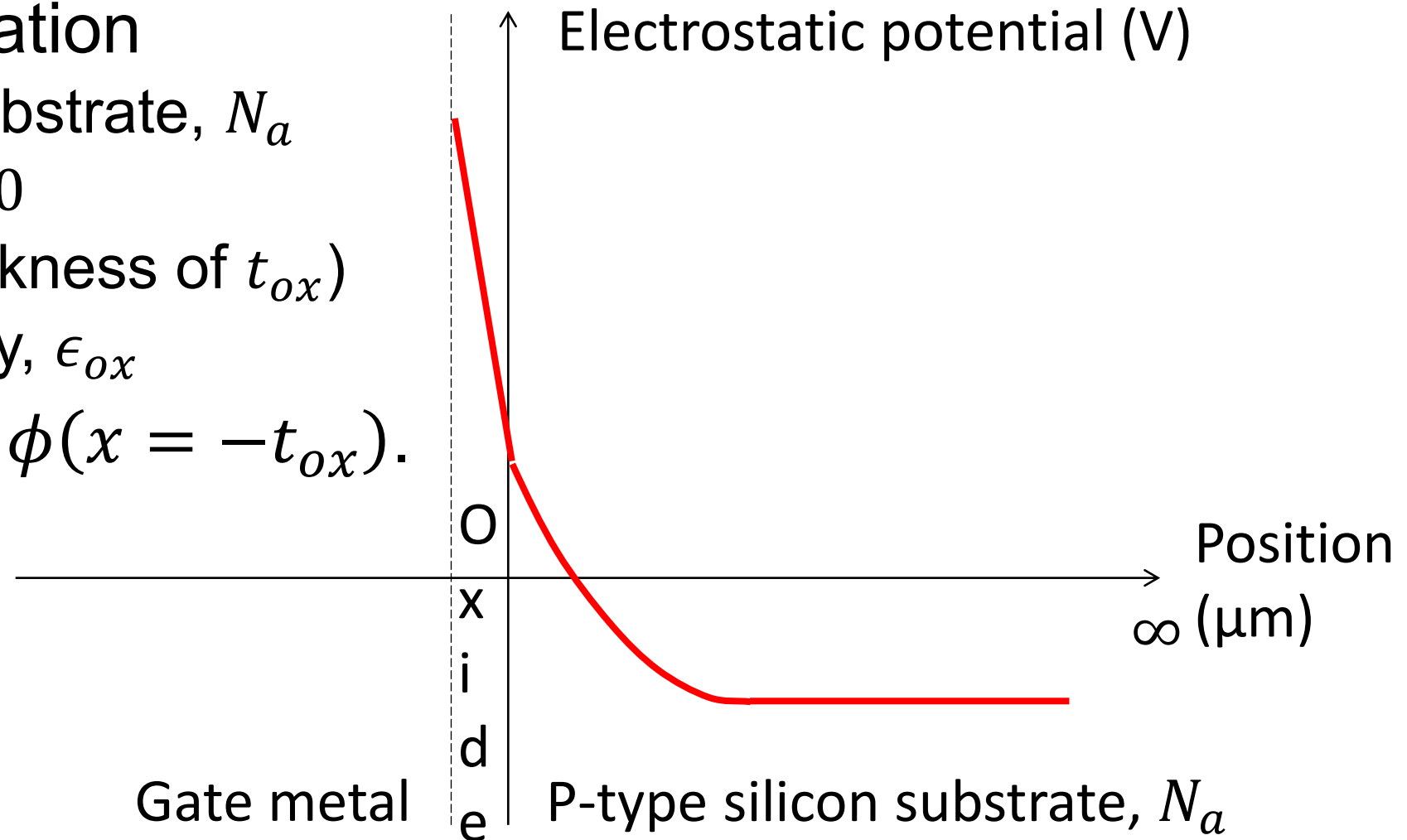
Department of Electrical Engineering and Computer Science

Gwangju Institute of Science and Technology (GIST)

L4

MOS capacitor (with oxide layer)

- Problem specification
 - P-type silicon substrate, N_a
 - Interface at $x = 0$
 - Oxide layer (thickness of t_{ox})
 - Oxide permittivity, ϵ_{ox}
- We must specify $\phi(x = -t_{ox})$.



Intrinsic Fermi level at gate

- Let us assume that $V_G = 0$ V.
 - For the gate metal, the workfunction (Φ_m) is known. The workfunction is the energy difference between the vacuum level and the Fermi level.
 - Therefore, when the workfunction is 4.3 eV, the vacuum level is located at 4.3 eV, because the Fermi level is the energy reference.
 - Moreover, the energy difference between the vacuum level and the intrinsic Fermi level of silicon is given. (About 4.61 eV)
 - Then, the intrinsic Fermi level of silicon (not of oxide) is located at -0.31 eV.

$$E_i = \Phi_m - \chi - (E_c - E_i)$$

GIST Lecture

Electron affinity

Electrostatic potential at gate

- From the intrinsic Fermi level (of silicon),
 - We can calculate the electrostatic potential as

$$-q\phi = E_i$$

- Simply speaking, we know $\phi(x = -t_{ox})$,

$$\phi(x = -t_{ox}) = -\frac{\Phi_m - \chi - (E_c - E_i)}{q}$$

- For a general V_G ,

$$\phi(x = -t_{ox}) = -\frac{\Phi_m - \chi - (E_c - E_i)}{q} + V_G$$

Flatband voltage, V_{fb}

- We are working on $\bar{\phi} = \phi - \phi_{\infty}$.

– Then,

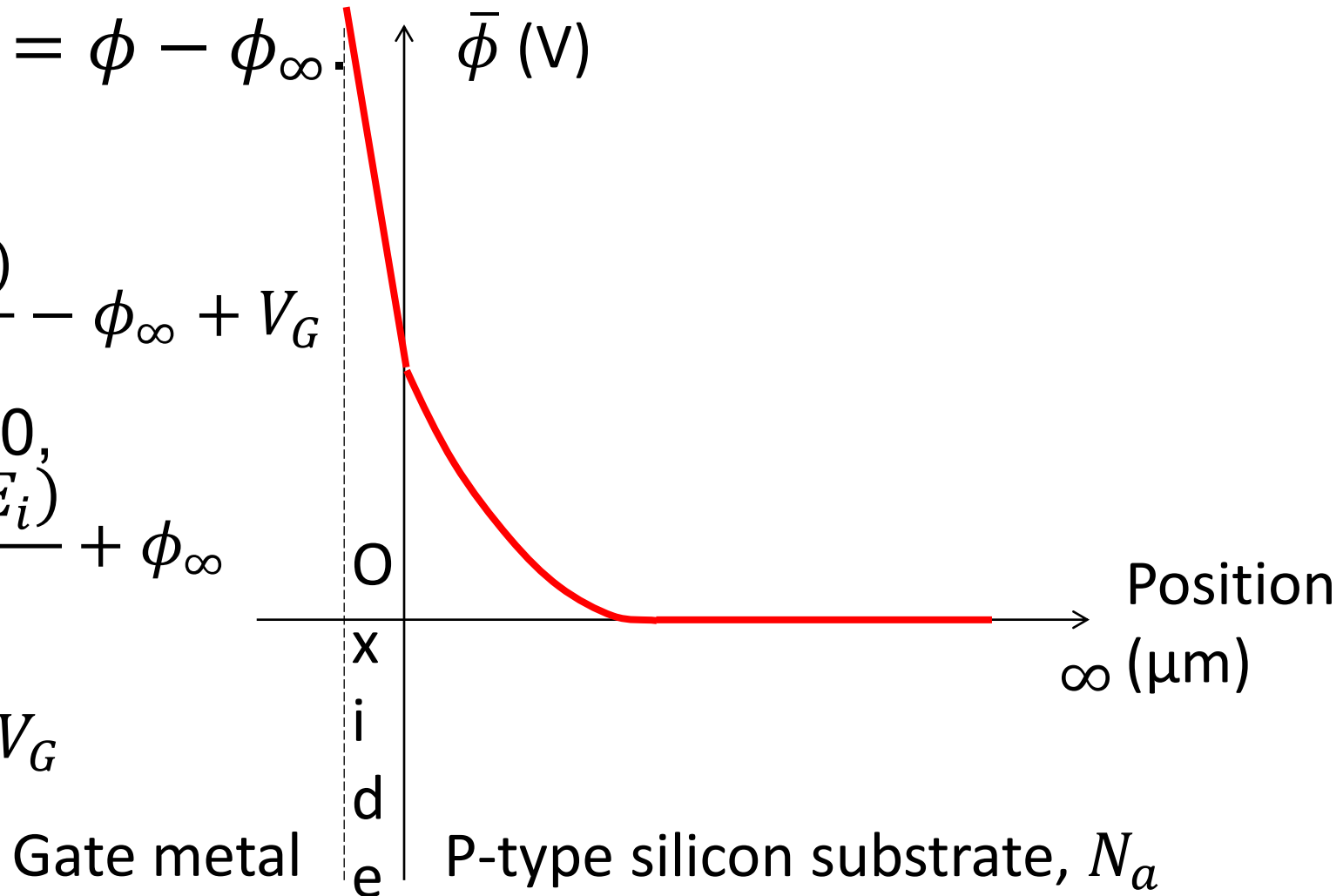
$$\bar{\phi}(x = -t_{ox}) = \frac{\Phi_m - \chi - (E_c - E_i)}{q} - \phi_{\infty} + V_G$$

– When $\bar{\phi}(x = -t_{ox}) = 0$,

$$V_{fb} = \frac{\Phi_m - \chi - (E_c - E_i)}{q} + \phi_{\infty}$$

– Therefore,

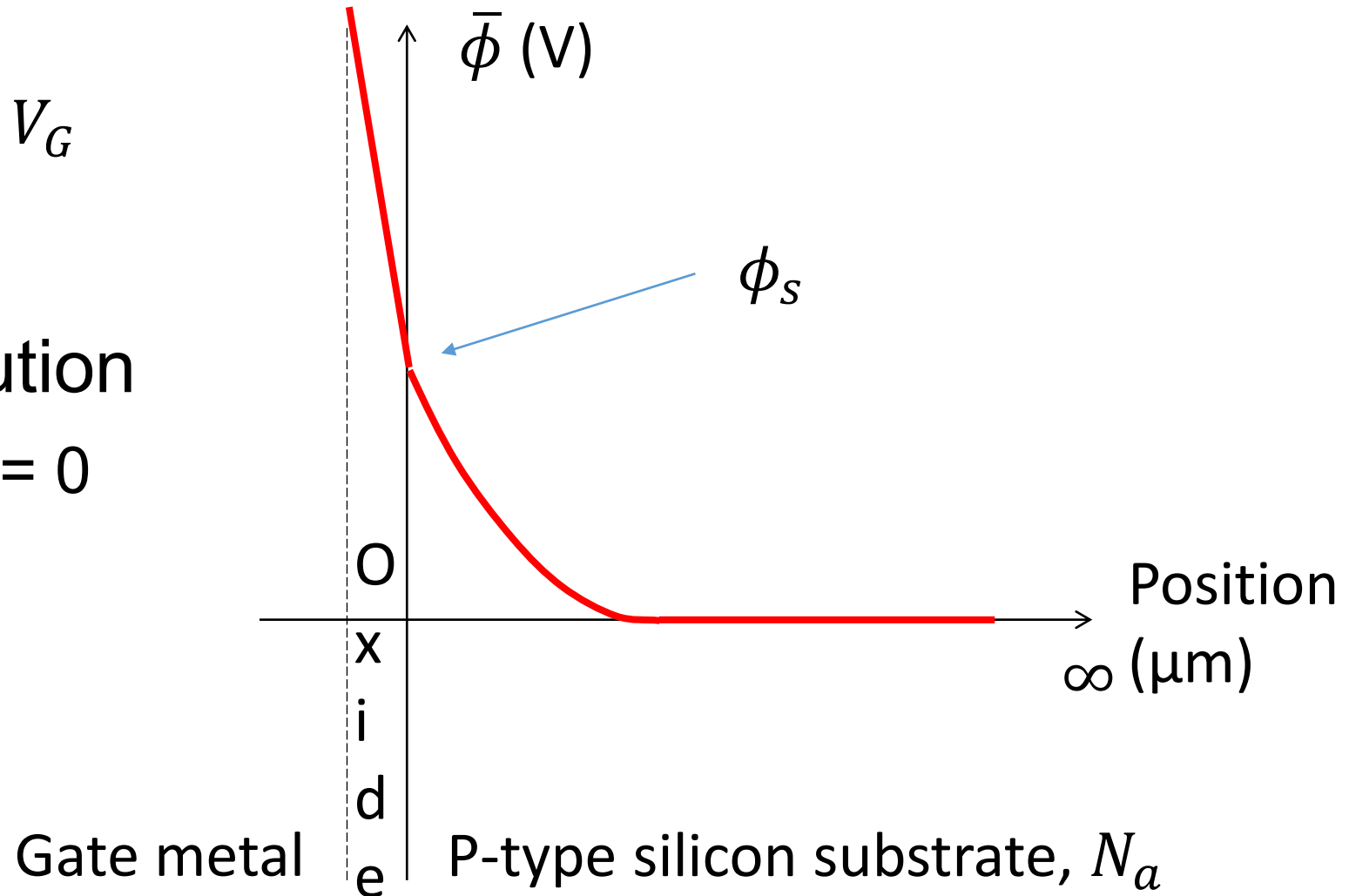
$$\bar{\phi}(x = -t_{ox}) = -V_{fb} + V_G$$



Values of $\bar{\phi}$

- Now we know that
 - $\bar{\phi}(x = -t_{ox}) = -V_{fb} + V_G$
 - $\bar{\phi}(x = 0) = \phi_s$
 - $\bar{\phi}(x = \infty) = 0$
- Ready to get the solution
 - Condition for $\frac{d\phi}{dx}$ at $x = 0$

$$\epsilon_{ox} \left. \frac{d\phi}{dx} \right|_{0^-} = \epsilon_{si} \left. \frac{d\phi}{dx} \right|_{0^+}$$



MOS equation

- With $\epsilon_{si} \frac{d\phi}{dx} \Big|_{0+} = Q_s$,

– The equation is simply written as

$$\epsilon_{ox} \frac{\phi_s - (-V_{fb} + V_G)}{t_{ox}} = Q_s$$

- With $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$,

$$\phi_s - (-V_{fb} + V_G) = \frac{Q_s}{C_{ox}}$$

$$V_G - V_{fb} = -\frac{Q_s}{C_{ox}} + \phi_s$$

(Of course, Q_s is a nonlinear function of ϕ_s .)

Expression for Q_s

- When $\phi_s > 0$, we have negative charges.
 - With the electron quasi-Fermi potential, ϕ_n ,

$$Q_s = - \sqrt{\frac{2\epsilon_{si}k_B T n_{int}}{q}} \times \left[\exp\left(-\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(-\frac{q\phi_s}{k_B T}\right) - 1 + \frac{q\phi_s}{k_B T} \right\} + \exp\left(\frac{q\phi_\infty}{k_B T}\right) \left\{ \exp\left(-\frac{q\phi_n}{k_B T}\right) \left(\exp\left(\frac{q\phi_s}{k_B T}\right) - 1 \right) - \frac{q\phi_s}{k_B T} \right\} \right]$$

(We assume that ϕ_n is a constant.)

Newton method

- It is straightforward to implement the Newton method.

- The equation for ϕ_s is

$$f(\phi_s) = -\frac{Q_s}{C_{ox}} + \phi_s - V_G + V_{fb} = 0$$

- The derivative is

$$\frac{df}{d\phi_s} = -\frac{1}{C_{ox}} \frac{dQ_s}{d\phi_s} + 1$$

(Of course, we know how to calculate $\frac{dQ_s}{d\phi_s}$.)

An example ($\phi_n = 0$)

- Let us calculate ϕ_s . For example, at $V_G = 1.0$ V?
 - Assume that Φ_m is 4.10 eV and χ is 4.05 eV.
 - Also, N_C at 300 K is $2.85665 \times 10^{19} \text{ cm}^{-3}$ and n_{int} at 300 K is $1.075 \times 10^{10} \text{ cm}^{-3}$. Therefore, $E_C - E_i$ is 0.561 eV.
 - Assume that the p-type doping concentration is 10^{17} cm^{-3} . Then, the flatband voltage, V_{fb} , is about -0.92581 V. When V_G is 1.0 V, $\bar{\phi}$ at gate is 1.92581 V.
 - The oxide thickness is 10 nm.
 - Then, ϕ_s is about 0.948 V.
- How about $V_G = 1.1$ V?

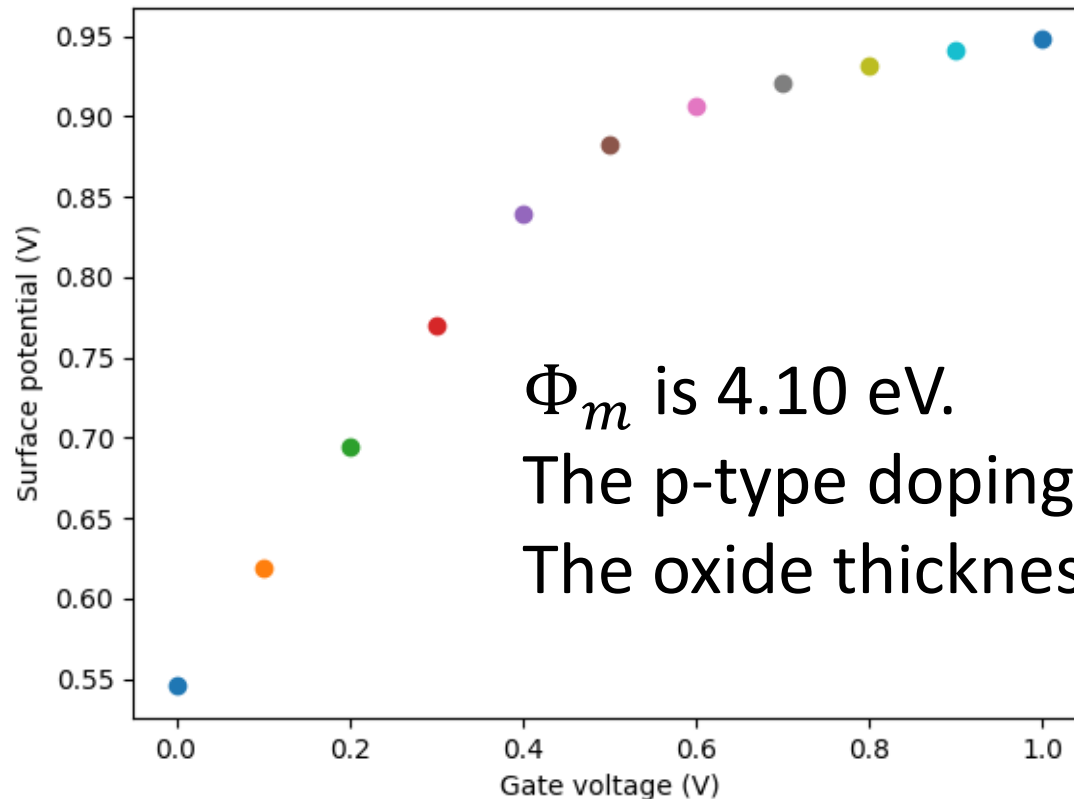
1.0	0.9479459310308174	0.0006777834886385816
1.0	0.9479408314730758	-5.0995577415125966e-06
1.0	0.947940831181903	-2.911728105350234e-10

 - Even in this case, ϕ_s is about 0.954 V. (Only slightly increased)

ϕ_s versus V_G ($\phi_n = 0$)

- Now, at a given V_G , we can calculate the surface potential.

– The total charge can be also obtained by $Q_s = \epsilon_{ox} \frac{\phi_s - (-V_{fb} + V_G)}{t_{ox}}$.



Φ_m is 4.10 eV.

The p-type doping concentration is 10^{17} cm^{-3} .

The oxide thickness is 10 nm.

Thank you!